AEGIS BANK

Risk Management Framework

Enterprise AI Decision & Knowledge Platform for Modern Banking

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Document Information

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Confidentiality Notice

This Risk Management Framework establishes the principles, processes, and governance structure used by Aegis Bank to identify, assess, monitor, and mitigate risks across all banking operations.

The framework ensures that financial, operational, cybersecurity, compliance, and technology risks are managed effectively.

Unauthorized disclosure of this document is prohibited.

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1. Purpose

The purpose of this framework is to establish a structured approach for managing risks that may affect Aegis Bank's:

- Financial stability.
- Customer trust.
- Regulatory compliance.

- Operational performance.
- Technology systems.
- AI-driven decision processes.

This framework enables proactive risk identification and effective decision-making.

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2. Scope

This framework applies to all Aegis Bank departments, including:

- Retail Banking.
- Corporate Banking.
- Compliance.
- Information Security.
- IT Operations.
- AI & Data Teams.
- Customer Support.
- Finance.
- Human Resources.
- Third-party vendors.

It covers risks related to:

- Financial activities.
- Customer operations.
- Digital banking.
- AI systems.
- Cybersecurity.
- Regulatory requirements.

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3. Risk Management Principles

Aegis Bank follows these core principles:

Risk Awareness

All employees must understand risks related to their responsibilities.

Proactive Management

Risks should be identified and addressed before causing business impact.

Data-Driven Decisions

Risk decisions should be supported by:

- Analytics.
- Monitoring.
- Reports.
- Historical data.

Continuous Improvement

Risk controls must be reviewed and improved regularly.

Accountability

Every identified risk must have:

- Owner.
- Assessment.
- Mitigation plan.
- Monitoring process.

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4. Risk Governance Structure

Aegis Bank maintains a three-line risk governance model.

First Line — Business Departments

Responsible for:

- Identifying operational risks.
- Following controls.
- Managing daily activities.

Examples:

- Banking Operations.
- Customer Support.
- Finance.

Second Line — Risk & Compliance

Responsible for:

- Risk monitoring.
- Policy development.
- Independent reviews.
- Compliance oversight.

Third Line — Internal Audit

Responsible for:

- Independent assessment.
- Control testing.
- Governance evaluation.

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5. Risk Management Process

Aegis Bank follows a continuous risk management lifecycle:

Identify Risk

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Analyze Risk

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Evaluate Impact

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Apply Controls

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Monitor Risk

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Review & Improve

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6. Risk Identification

Risks may be identified through:

- Internal audits.
- Security monitoring.
- Customer complaints.
- Regulatory updates.
- System reviews.
- Employee reports.
- Business assessments.

Risk Sources

Common risk sources include:

Internal Sources

- Process failures.
- Employee mistakes.
- System errors.
- Security weaknesses.

External Sources

- Cyber attacks.
- Regulatory changes.
- Market changes.
- Vendor failures.

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7. Risk Assessment Methodology

Aegis Bank evaluates risks using:

Probability

Likelihood of risk occurring.

Scale:

Level	Description
Low	Rare occurrence
Medium	Possible occurrence
High	Likely occurrence

Impact

Potential business damage.

Scale:

Level	Description
Low	Limited impact

Level	Description
Medium	Moderate impact
High	Severe impact

Risk Score Calculation

Risk Score:

Probability × Impact

Example:

A cybersecurity attack:

Probability = High (3)

Impact = High (3)

Risk Score:

3 × 3 = 9

Risk Level:

➡ Critical

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8. Risk Classification

Aegis Bank classifies risks into categories:

Risk Category	Examples
Credit Risk	Loan defaults
Operational Risk	Process failures
Cyber Risk	Data breaches
Compliance Risk	Regulatory violations
Market Risk	Financial changes

Risk Category	Examples
Technology Risk	System failures
AI Risk	Model errors

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9. Risk Mitigation Strategies

Aegis Bank uses four primary risk treatment methods:

Avoid

Remove activities creating unacceptable risk.

Example:

Blocking unauthorized applications.

Reduce

Apply controls to lower risk.

Example:

Using MFA for banking systems.

Transfer

Move risk responsibility.

Example:

Cyber insurance or vendor agreements.

Accept

Accept low-level risks after evaluation.

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10. Credit Risk Management

Credit risk is the possibility that customers may fail to meet financial obligations.

Controls include:

- Credit scoring.
- Income verification.
- Loan assessment.
- Risk-based approvals.
- Portfolio monitoring.

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11. Operational Risk Management

Operational risk arises from:

- Human errors.
- Process failures.
- System issues.
- Internal incidents.

Controls include:

- Standard operating procedures.
- Employee training.
- Process monitoring.
- Backup procedures.

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12. Cybersecurity Risk Management

Cyber risks include:

- Unauthorized access.
- Malware.
- Data breaches.

- Phishing attacks.

Controls include:

- Security monitoring.
- Vulnerability management.
- MFA.
- Encryption.
- Incident response procedures.

Related Document:

AB-SEC-004 — Cyber Security Incident Response Plan

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13. Compliance Risk Management

Compliance risks include:

- GDPR violations.
- AML failures.
- Regulatory breaches.

Controls include:

- Compliance monitoring.
- Internal audits.
- Regulatory reviews.
- Employee training.

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14. AI Risk Management

Aegis Bank manages risks related to AI systems including:

- Model inaccuracies.
- Bias.
- Data leakage.

- Hallucinations.
- Unauthorized AI usage.

AI Risk Controls

Aegis AI OS follows:

- Human oversight.
- Model evaluation.
- Source verification.
- Audit logging.
- Access control.

AI responses must be evaluated for:

- Accuracy.
- Reliability.
- Security impact.

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15. Third-Party Risk Management

Third-party vendors are assessed before approval.

Assessment includes:

- Security capability.
- Privacy compliance.
- Business continuity.
- Financial stability.
- Regulatory requirements.

16. Risk Monitoring & Reporting

Aegis Bank continuously monitors identified risks to ensure that mitigation strategies remain effective.

Risk monitoring activities include:

- Reviewing risk indicators.
- Tracking mitigation progress.
- Evaluating control effectiveness.
- Reporting emerging risks.
- Updating risk assessments.

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17. Key Risk Indicators (KRIs)

Key Risk Indicators (KRIs) are measurable values used to identify increasing risk levels before major impact occurs.

Examples of KRIs:

Risk Area	Key Risk Indicator
Cybersecurity	Number of security incidents
Credit Risk	Loan default percentage
Compliance	Number of regulatory violations
Operations	Process failure rate
AI Systems	Model accuracy degradation
Vendors	Third-party security issues

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18. Risk Reporting Structure

Risk reports are generated regularly and shared with relevant stakeholders.

Daily Monitoring

Performed by operational teams.

Includes:

- System alerts.
- Security events.
- Transaction anomalies.

Monthly Risk Reports

Prepared by Risk Management Department.

Includes:

- Current risks.
- Risk trends.
- Control effectiveness.
- Outstanding actions.

Quarterly Risk Review

Reviewed by:

- Executive Management.
- Risk Committee.
- Board Representatives.

Includes:

- Strategic risks.
- Emerging threats.
- Business impact.

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19. Risk Escalation Process

High-impact risks must be escalated immediately.

Risk escalation levels:

Risk Level	Action
Low	Department monitoring
Medium	Manager review
High	Risk Department involvement
Critical	Executive escalation

Critical Risk Examples

- Major customer data breach.
- Banking system outage.
- Regulatory violation.
- AI decision failure affecting customers.
- Large-scale fraud event.

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20. Risk Management Dashboard

Aegis Bank maintains a centralized risk dashboard.

The dashboard provides visibility into:

- Active risks.
- Risk severity.
- Department ownership.
- Mitigation progress.
- Historical trends.

Dashboard Metrics

Metric	Description
Total Risks	Number of identified risks
Critical Risks	Highest priority risks
Open Actions	Pending mitigation activities
Resolved Risks	Completed risk treatments
Risk Trend	Increasing or decreasing exposure

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21. AI Risk Assessment Framework

Because Aegis Bank uses AI technologies, AI risks require dedicated evaluation.

AI Risk Categories

1. Model Accuracy Risk

Risk:

AI produces incorrect recommendations.

Controls:

- Model testing.
- Performance evaluation.
- Human review.

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2. Hallucination Risk

Risk:

AI generates inaccurate information.

Controls:

- Retrieval-Augmented Generation (RAG).
- Source verification.

- Confidence scoring.

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3. Data Privacy Risk

Risk:

AI exposes sensitive customer information.

Controls:

- Access restrictions.
- Data classification.
- Encryption.

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4. Bias Risk

Risk:

AI decisions may unfairly impact users.

Controls:

- Bias testing.
- Regular evaluation.
- Human oversight.

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5. Security Risk

Risk:

Attackers manipulate AI systems.

Controls:

- Prompt security.
- Monitoring.
- Access control.

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22. Aegis AI OS Risk Controls

Aegis AI OS follows enterprise AI governance standards.

The system provides:

- Document-grounded responses.
- Source citations.
- Model comparison.
- Hallucination evaluation.
- LangFuse tracing.
- User activity monitoring.

AI Decision Flow

Employee Question

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Access Verification

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Document Retrieval (RAG)

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Multiple LLM Comparison

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Response Evaluation

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Final Answer + Sources

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Logging & Monitoring

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23. Business Continuity Risk Management

Aegis Bank prepares for operational disruptions.

Risks include:

- Server failures.
- Natural disasters.
- Cyber attacks.
- Infrastructure outages.

Controls include:

- Backup systems.
- Disaster recovery plans.
- Alternative processing locations.
- Emergency procedures.

Related Documents:

- AB-DR-001 Backup Strategy
- AB-DR-002 Disaster Recovery Plan

- AB-DR-003 Business Continuity Plan

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24. Banking Risk Scenarios

Scenario 1 — Loan Default Increase

Situation:

Loan default rates increase significantly.

Risk Category:

➡️ Credit Risk

Response:

- Review lending criteria.
- Analyze customer risk profiles.
- Adjust approval controls.
- Monitor portfolio.

Scenario 2 — Customer Data Breach

Situation:

Unauthorized access to customer records occurs.

Risk Category:

➡️ Cybersecurity Risk

Response:

- Activate Incident Response Plan.
- Block unauthorized access.
- Investigate affected systems.
- Apply security improvements.

Scenario 3 — AI Provides Incorrect Policy Answer

Situation:

Aegis AI OS provides an inaccurate response.

Risk Category:

➡️ AI Model Risk

Response:

- Review retrieved documents.
- Analyze model output.
- Update knowledge sources.
- Improve evaluation process.

Scenario 4 — Vendor Security Failure

Situation:

A third-party provider experiences a security incident.

Risk Category:

➡️ Third-Party Risk

Response:

- Review vendor impact.
- Restrict access.
- Conduct investigation.
- Apply corrective actions.

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25. Risk Management Responsibilities

Executive Management

Responsible for:

- Approving risk strategy.

- Reviewing major risks.
- Supporting risk culture.

Chief Risk Officer (CRO)

Responsible for:

- Managing risk framework.
- Reporting risk exposure.
- Coordinating risk activities.

Department Managers

Responsible for:

- Identifying operational risks.
- Implementing controls.
- Reporting issues.

Employees

Responsible for:

- Following policies.
- Reporting risks.
- Protecting company assets.

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26. Frequently Asked Questions (FAQs)

Q1. Why does Aegis Bank need a Risk Management Framework?

It helps identify, evaluate, and control risks that may impact banking operations, customers, and regulatory compliance.

Q2. Who owns risk management at Aegis Bank?

Risk management is a shared responsibility between executives, departments, employees, and the Risk Management Team.

Q3. How are critical risks handled?

Critical risks are immediately escalated to executive management and require urgent mitigation actions.

Q4. How does Aegis Bank manage AI risks?

Through RAG grounding, model evaluation, source verification, monitoring, and human oversight.

Q5. What happens when a risk is identified?

The risk is recorded, assessed, assigned an owner, and managed through appropriate controls.

Q6. Are all risks eliminated?

No. The goal is to identify, reduce, monitor, and manage risks within acceptable levels.

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27. Related Documents

This framework should be read together with:

- GDPR & Privacy Policy (AB-COMP-001)
- Customer Privacy Policy (AB-COMP-002)
- Internal Audit Manual (AB-COMP-004)
- Regulatory Compliance Manual (AB-COMP-005)
- Cyber Security Policy (AB-SEC-003)
- Data Classification Policy (AB-SEC-005)
- AI Risk Management Policy (AB-AI-004)

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28. Revision History

Version	Date	Description	Author
v1.0	February 20, 2026	Initial Risk Management Framework Release	Enterprise Risk Management
v1.1	May 25, 2026	Added AI Risk Assessment and Monitoring Controls	Chief Risk Officer
v1.2	**July 24, 2026**	**Enhanced Risk Dashboard, AI Governance, and Enterprise Risk Controls**	**Enterprise Risk Management Department**

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29. Document Approval

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End of Document

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